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2014.0 RANGE ROVER (LG), 303-06

STARTING SYSTEM - V8 N/A 5.0L PETROL/V8 S/C 5.0L PETROL

PRINCIPLES OF OPERATION

For a detailed description of the starting system and operation, refer to the relevant Description and Operation section in the Workshop Manual.

REFER to: Starting System (303-06 Starting System - V8 5.0L Petrol/V8 S/C 5.0L Petrol, Description and Operation).

INSPECTION AND VERIFICATION



CAUTION:

Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Gear selector lever cable adjustment (vehicles with automatic transmission) ▪ Starter motor ▪ Engine (turns freely)	▪ Battery ▪ Fuses ▪ Starter relay ▪ Wiring harness(es) ▪ Damaged, loose or corroded connectors ▪ Ignition switch ▪ Generator ▪ Engine Control Module (ECM) ▪ Transmission Control Module (TCM)

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.
1. Check DDW for open campaigns. Refer to the corresponding bulletins and SSM's which may be valid for the specific customer complaint and carry out the recommendations as needed.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
The engine does not crank (starter motor does not turn)	▪ Gear selector not in P or N position (vehicles with automatic transmission) ▪ Battery ▪ Starter relay ▪ Output circuit: high resistance ▪ Output circuit: short circuit to power ▪ Invalid key code received by Central Junction Box (CJB) ▪ Harness /Connectors ▪ Starter motor ▪ Ignition switch ▪ Generator ▪ Engine seized	Make sure the gear selector is in the P or N position and correctly adjusted. Check the battery condition and state of charge. Check for DTCs indicating an immobilizer fault. Check the starter motor relay, ignition switch and generator circuits. Refer to the electrical guides. Check that the engine turns freely.
The engine does not crank	▪ Starter motor fitment ▪ Starter motor	Check the starter motor fitment (fasteners tight, starter motor square to engine, etc). Check the flywheel/drive plate ring gear teeth for damage, foreign objects, etc.

(starter motor does turn)	▪ Flywheel /Drive plate ring gear	
Engine cranks too slowly	▪ Battery ▪ Harness /Connectors ▪ Starter motor ▪ Oil grade	Check the battery condition and state of charge. Check the starter motor circuits. Refer to the electrical guides. Check the engine oil grade and condition.
Engine cranks too fast	▪ Low engine compression	Check the engine condition and compressions.
Excessive starter motor noise	▪ Starter motor ▪ Flywheel /Drive plate ring gear ▪ Starter motor installation /casing	Check the starter motor fitment (fasteners tight, motor square to engine, etc). Check the starter motor casing condition. Check the flywheel/drive plate ring gear teeth for damage, foreign objects, etc.
ECO (stop /start) system inoperative	▪ ECO system inhibited ▪ ECO system fault	 NOTE: "ECO STOP/START FAULT" will be displayed in the instrument cluster message centre when the system is inoperative due to a fault. ▪ Refer to the ECO System Inhibits table for possible inhibit conditions ▪ ECO system inhibited due to owner error: Explain /demonstrate the ECO system operation to the owner ▪ ECO system inhibited due to battery condition: Using the manufacturer approved diagnostic system, perform the Power Supply Service Mode Diagnostic routine ▪ Using the manufacturer approved diagnostic system, check the Engine Control Module (ECM) for related DTCs and refer to the relevant DTC index ▪ Automatic/ECO stop/start system inhibited due to a DTC clear performed and a drive cycle not completed: Road test the vehicle between 48-80 km/h (30-50 mph) for 2 to 3 minutes, allow the engine to idle, auto stop/start will now activate

ECO SYSTEM INHIBITS

The inhibit conditions listed below will not set Diagnostic Trouble Codes (DTCs) but will set flags which can be viewed using the manufacturer approved diagnostic system -

Datalogger / Engine Control Module (ECM). Normal ECO system operation will resume when an inhibit condition is no longer present, but the battery related inhibits may persist for some time.

NOTES:

- Performing a DTC clear on a petrol (gas) engine vehicle will inhibit the auto stop/start system.
- Do not reset the Battery Monitoring System (BMS) to rectify battery condition inhibits. This may temporarily restore ECO system operation but will lead to a recurrence of the fault.

INHIBITS

- ECO switch has been pressed by driver
- DTC clear performed
- Transmission not in neutral
- Clutch pedal not fully released
- Accelerator pedal not released
- Hood open detected
- Driver door detected open
- Driver seatbelt not fastened
- Hill Descent Control (HDC) is switched on
- Terrain response special programme selected
- Climate control demand exceeds calibrated threshold
- Heated windshield operating
- Trailer connected
- Brake booster vacuum below threshold
- Engine coolant temperature below threshold
- Engine oil temperature below threshold
- Catalytic converter outside either pre or post calibration range
- Diesel Particulate Filter (DPF) active regeneration in progress (diesel engines only)
- Battery state of charge low
- Battery cold cranking capability below threshold
- Ambient temperature above 40°C
- Ambient temperature below 0°C

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: Diagnostic Trouble Code (DTC) Index - V8 S /C 5.0L Petrol, DTC: Engine Control Module (ECM) (100-00 General Information, Description and Operation).